

Module 0: Project Overview and Phase Map

From Radiation-Induced Defects to Device Degradation in Silicon

Purpose of this overview

This document is the map of the whole project. The later modules go deep into the physics, derivations, and assumptions inside each phase. This overview answers a simpler question first: *how do all of the pieces fit together?*

The project tells one continuous story:

incident radiation $\rightarrow$ transport in silicon $\rightarrow$ displacement-like events $\rightarrow$ defect population $\rightarrow$ semiconductor property changes $\rightarrow$ device degradation $\rightarrow$ cryogenic persistence and slower recovery.

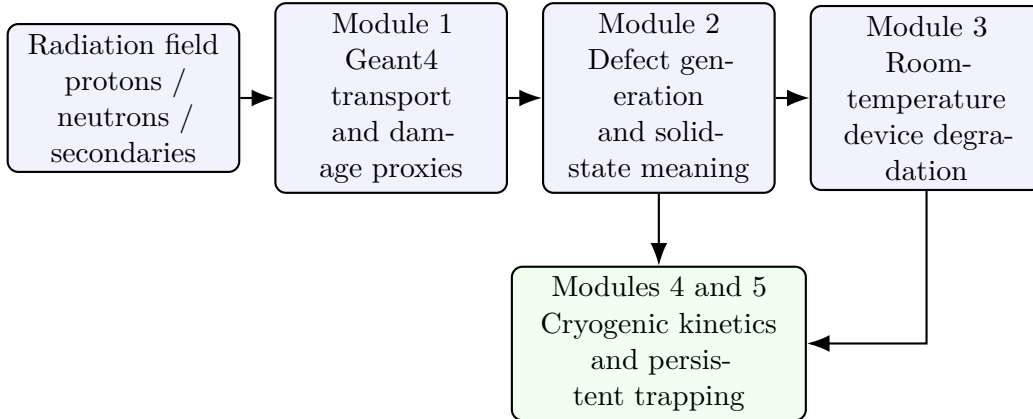
A useful analogy is a winter road after a storm. The storm itself is the radiation field. Tire tracks and impacts in the snow are the transport and collision history. Cracks and potholes are the defects. The way the road drives afterward is the device-level degradation. If the temperature later drops far below freezing, the road damage can become locked in and much slower to heal. That final low-temperature behavior is the cryogenic extension of this project.

The five-module structure

The project is organized into five modules.

Phase	Module title	Main physics question	Main output
0	Project overview and phase map	How do all parts connect into one causal chain?	Global roadmap and workflow
1	Geant4 transport and damage proxies	What happens when radiation enters silicon?	Depth deposition, event deposition, recoil-like proxies
2	Defect generation and solid-state interpretation	How do interaction histories become defects and traps?	Defect-density model and interpretation of lattice damage
3	Room-temperature device degradation	How do defects alter lifetime, diffusion length, leakage, and charge collection?	Device-level degradation trends
4	Cryogenic defect kinetics	How do migration, emission, and annealing change as temperature falls?	Temperature-dependent kinetics and freeze-out trends
5	Persistent trapping and slow recovery	What cryogenic device consequence follows from frozen kinetics?	Slow recovery and radiation-history persistence

Visual pipeline



The key structural idea is that Modules 1–3 form the main room-temperature chain, while Modules 4–5 are an extension that asks how the defect landscape changes when the silicon is cooled toward the cryogenic limit.

What each module contributes

Module 1: Geant4 transport and damage proxies

This module is the front end. It answers: what interactions occur inside silicon before we ever talk about defects? Geant4 provides

- event-by-event deposited energy,

- depth-dependent deposited energy,
- recoil-like ion secondaries as a proxy for strong lattice perturbation.

The central caution is that Geant4 does *not* directly give a final, annealed defect chemistry. It gives the collision history from which we construct a physically motivated bridge to defect formation.

Module 2: Defect generation and solid-state interpretation

This module converts transport outputs into a defect-centered picture. The main bridge is a first-order relation such as

$$N_D \propto \Phi \quad \text{or} \quad N_D(z) \propto D_{\text{proxy}}(z), \quad (1)$$

where N_D is defect density, Φ is fluence, and $D_{\text{proxy}}(z)$ is a damage proxy from Module 1.

Physically, this is the stage where vacancies, interstitials, and defect complexes enter the story. It is where the project moves from radiation transport into solid-state physics.

Module 3: Room-temperature device degradation

This module connects defects to measurable semiconductor degradation. Representative relations are

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_0} + K_{\tau} N_D, \quad (2)$$

$$L = \sqrt{D \tau_{\text{eff}}}, \quad (3)$$

$$\Delta I = \alpha \Phi V. \quad (4)$$

These equations connect defects to lifetime reduction, diffusion-length reduction, and leakage-current increase.

A good analogy is traffic flow in a city. Defects are like intersections blocked by construction. Carriers have a harder time moving efficiently, more of them get delayed or lost, and the overall traffic quality degrades.

Module 4: Cryogenic defect kinetics

This module asks how the *behavior* of existing defects changes with temperature. It focuses on three thermally activated processes:

$$D_{\text{def}}(T) = D_0 \exp\left(-\frac{E_m}{k_B T}\right), \quad (5)$$

$$e(T) = e_0 \exp\left(-\frac{E_a}{k_B T}\right), \quad (6)$$

$$R_{\text{ann}}(T) = R_0 \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right). \quad (7)$$

As temperature drops, migration, emission, and annealing all slow dramatically. The main message is not that defects disappear in the cold. The message is that many of their *kinetics* freeze out.

Module 5: Persistent trapping and slow recovery

This module turns the cryogenic kinetics into one concrete device consequence. A simple recovery model is

$$Q_{\text{trap}}(t, T) = Q_0 \exp\left(-\frac{t}{\tau_{\text{rec}}(T)}\right), \quad (8)$$

with

$$\tau_{\text{rec}}(T) = \tau_{00} \exp\left(\frac{E_r}{k_B T}\right). \quad (9)$$

The interpretation is intuitive: at low temperature, the recovery clock can become so slow that the device effectively remembers part of its radiation history over operational timescales.

Development order versus run order

These two ideas are different, and separating them removes confusion.

Development order

Development order means the order in which the project was architected and expanded:

1. build the room-temperature physics chain,
2. connect Geant4 outputs to that chain,
3. add cryogenic defect kinetics,
4. add persistent trapping as the cryogenic device consequence.

This order is about project design and scope growth.

Practical run order

Practical run order means the easiest order for the user to execute and understand the scripts:

1. run the self-contained room-temperature MATLAB pipeline,
2. run the cryogenic kinetics extension,
3. run the persistent trapping extension,
4. only after Geant4 CSV files exist, run the Geant4-import MATLAB pipeline.

This order is about learning and debugging without being blocked by Geant4 setup.

How the code layers map onto the physics

Code component	Physics role	Typical use
Geant4 application	Radiation transport and collision history	Produce damage proxies and spatial interaction data
main_full_pipeline.m	Core room-temperature degradation model	Learn the main defect-to-device chain without Geant4 input
main_import_geant4_	Geant4-coupled degradation model	Map Geant4 output into depth-dependent or fluence-based damage input
main_cryogenic_extensions	Cryogenic kinetics model	Study migration, trap emission, and annealing freeze-out
main_persistent_trapping	Cryogenic device consequence model	Show slow recovery and radiation-history persistence

What the user should remember

- This is one project, not five unrelated documents.
- Modules 1–3 form the main radiation-to-device chain.
- Modules 4–5 are a cryogenic extension of that same chain.
- Development order and run order are different on purpose.
- The whole project becomes strongest when discussed as one causal story rather than as isolated equations.